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Narula Institute of Technology
An Autonomous Institute under MAKAUT
2023

END SEMESTER EXAMINATION - ODD 2023

M(IT)101 - Engineering Mathematics - I

TIME ALLOTTED: 3Hours

FULL MARKS: 70

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A

(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. Sum of two homogeneous functions is homogeneous if (1) CO1 BL1
- a) they have same degree of homogeneity
- b) they are both continuous
- c) they are both derivable
- d) none of these

2. The eigen values of the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$ are (1) CO2 BL2

- a) 0,0,1
- b) 1,2,3
- c) 2,3,6
- d) None of these

3. If $f_x(a,b)=f_y(a,b)=0$, then (a,b) is (1) CO1 BL1
- a) saddle point
- b) critical point
- c) isolated point
- d) none of these

4. The value of $\int_0^2 \int_0^2 \int_0^2 xyz \, dx dy dz$ is (1) CO2 BL2

- a) 1/2
- b) 1/8
- c) 8
- d) 16

5. $\int_0^1 \int_y^{\sqrt{y}} dx dy =$ (1) CO2 BL2

- a) 1/2
- b) 1/6
- c) 2/3
- d) 4/3

6. $u(x,y) = \frac{x^3 + x^2y}{\sqrt{x} + \sqrt{y}}$ is a homogeneous function of degree ____

(1) CO1 BL1

- a) 3/2
- b) 5/2
- c) 1/2
- d) None of these

7. The infinite series $\sum \frac{1}{(1+2n)^{100}}$ is

(1) CO3 BL1

- a) Convergent
- b) Divergent
- c) Oscillatory
- d) nothing to be said

8. If $f(x,y)$ is a homogeneous function of degree 3 then $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = kf$. What is the value of k?

(1) CO2 BL1

- a) 3
- b) 2
- c) 0
- d) 5

9. Two eigen values of a matrix $A = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$ are 2 and -2. The third eigen value is

(1) CO1 BL2

- a) 1
- b) 0
- c) 3
- d) 2

10. The critical point of the two functions y is (a) (1,1)

(1) CO4 BL1

- (b) (1,-1)
- (c) (-1,1)
- (d) (0,0)

11. $\int_0^1 \int_y^{\sqrt{y}} dx dy =$ (1) CO2 BL1
 a) 1/2
 b) 1/6
 c) 2/3
 d) 4/3
12. The characteristic equation of a matrix A is $X^3 + 3X^2 + 5X + 9 = 0$, then the determinant of the matrix is (1) CO2 BL1
 a. 7 b. 5 c. 6 d. 9

Group B
(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

13. Test the convergence of the power series $\frac{x}{1} + \frac{1}{2} \cdot \frac{x^2}{3} + \frac{1.3}{2.4} \frac{x^5}{5} + \frac{1.31.5}{214.6} \cdot \frac{x^7}{7} + \dots \infty$ (5) CO3 BL3
14. If $u = v^n e^{-\frac{r^2}{4v}}$ find n which satisfy the following $\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial u}{\partial r}) = \frac{\partial u}{\partial v}$ (5) CO3 BL3
15. Apply Cayley-Hamilton theorem for the matrix $\begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}$ to evaluate A^{-1} . (5) CO3 BL3
16. Find all Eigen values and one Eigen vector corresponding to one Eigen value of the given matrix. (5) CO4 BL2

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 2 & -1 & 0 \\ 5 & 9 & -1 \end{bmatrix}$$
17. Verify Rolle's theorem for the function $f(x) = x^2 - 5x + 6$ in $[2, 3]$ (5) CO4 BL4

Group C
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

- 18a. Investigate the values of λ and μ , for which the system of equations
- $$\begin{aligned}x + y + z &= 6 \\x + 2y + 3z &= 10 \\x + 2y + \lambda z &= \mu\end{aligned}$$
- may have
- unique solution
 - infinite number of solutions
 - no solution
- (6) CO4 BL4
- 18b. Reduce the following matrix $\begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & 1 & 2 \end{pmatrix}$ to Echelon form and hence find its rank.
- (3) CO2 BL2
- 18c. Apply elementary row operations to find A^{-1} of the following matrix: $\begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}$
- (6) CO3 BL3
19. a)
- (15) CO3 BL1
- If $f(x, y) = xy \frac{x^2 + 5y^2}{x^2 + y^2}, x^2 + y^2 \neq 0$
 $= 0, x=0, y=0$
 Prove that $f_{xy}(0,0) \neq f_{yx}(0,0)$
- b) If $u = \tan^{-1} \frac{x^2 + y^2}{x - y}$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \sin 2u$ (8+7)
20. For what value of λ , and μ do the system of equations
- $$\begin{aligned}x + y + z &= 6 \\x + 2y + 3z &= 10 \\x + 2y + \lambda z &= \mu\end{aligned}$$
- Have (i) No solution (ii) unique solution (iii) more than one solution.
- Hence solve for each cases.
- (15) CO2 BL1
21. 1) Evaluate $\iint (x^2 + y^2) dx dy$ over the region bounded by $xy=1, y=0, y=x$ and $x=2$.
- (15) CO4 BL1
- 2) Evaluate $\iiint z^2 dx dy dz$ over the region bounded by $z \geq 0, x^2 + y^2 + z^2 \leq a^2$ (6+9)

22. a) Test the convergence of the series

(15) CO3 BL1

$$1 + \frac{3}{7}x + \frac{3.6}{7.10}x^2 + \frac{3.6.9}{7.10.13}x^3 + \dots \dots \dots (x > 0)$$

b) Reduce the matrix in normal form and find its rank $\begin{pmatrix} 2 & 2 & 2 \\ 1 & 2 & 1 \\ 3 & 4 & 3 \end{pmatrix}$

(7+7+1)

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